BİLKENT UNIVERSITY

ENGINEERING FACULTY

DEPARTMENT OF COMPUTER ENGINEERING

CS353

Project Proposal

Report

Group 4

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1. Introduction

Shipping companies are essential for the transportation of the packages among those having a demand to send belongings to another person. However, a data management system is necessary to keep track of the flow of goods since it would be impossible to transport goods without knowing the necessary data. Therefore, providing a web application that assures a management system of keeping track of necessary data such as customer information, branch information, employee information, or even the distance information between the branches of a shipping company. The aim of this system is to simplify and improve the efficiency of the process of transporting various goods among people and locations with the necessary data. From the perspective of such a concept, a system that includes various panels and data management systems to ease the process of the delivery of the goods will be proposed as a project.

2. Description

This platform aims to make the shipping process easier. It enables users to send packages to a destination by either calling a courier to their own locations or submitting the packages to any working shipping branch in person. In the first case, the user fills out a form through the web application to call the courier and enters necessary data into the system such as name, surname, his/her address, destination address, the weight of the package for the price calculation, and contact number. After the data entering process, the user will enter his/her card information for the payment. According to the weight of the package, the cost of shipping is calculated through a pricing policy that is decided by the

company and the cost is added to a specific starting fee. The total price is withdrawn from the customer card. Then, the courier comes to the address to pick up the package and takes the package to the shipping branch. As a result, a package id is generated by the system and shown to the user. Therefore, the whole process is essential to get a package id for keeping track of the package and order. In the second case, the user submits the package to the branch in person.

In the branch, package managers direct the shipping process. They create a shipping request for each face-to-face order and provide the package id to the customer. After that, they assign the package to a shipper to deliver it to the destination branch. Each vehicle — courier vehicle or shipper vehicle — has a carrying capacity. Therefore, the shipping process needs to be managed in a planned way by the package manager to send the packages successfully by assigning packages into available vehicles. After delivering packages to the destination branches, packages are sent to the destination addresses via couriers. If the submitted package is malformed or the package is lost and can't be submitted to the destination address, the user is allowed to file a report to inform the company about the situation from the website. If the user is right about the informed situation, the shipping price is transferred into the customer's card so as not to allow unjust treatment. If the package is delivered without any problem to the destination address but is not received by the destination address host, the package is brought back to the destination branch by the courier and is kept in the branch for a specific length of time. After a certain time limit, the holdout packages will be resent to the customer

address with the same price. In addition, users can track whether their packages are submitted, on the road, or being held out in the destination branch from the platform.

Also, the platform contains an authorized user type which is the administrator. The administrator can open a new branch, close a branch, recruit a new employee to the company or fire an existing employee.

2.1 Database as a System

This shipping company data management system requires data to be stored, updated, and accessed. Customer, administrator, package manager, shipper and courier information, package properties, customer card information, shipping requests, reported files' records, and package delivery state information needs to be stored in a database. Database usage is inevitable for that kind of system because different users interact with each other frequently, and access information about themselves and packages. Also, since the data in the system are related to each other and need to be stored in rows and columns, the relational database model is selected for the shipping company system.

3. Requirements

3.1 Functional Requirements

There will be functional requirements for each user type (user, customers, employees, couriers, package managers, and shippers).

3.1.1 User

- Users can search packages according to package ids.

- Users can see the place and the situation of the packages from the system.

3.1.1.1 Customers

- Customers can call couriers to their own location by filling a form with the necessary information on the website.
- Customers can give packages to the branches in person.
- Customers can file a report in case of malformed or lost packages.

3.1.1.2 Employees

- Employees are able to search for other employees from the system
- Employees can see and update the situation of the packages from the system.
- Employees can log in and log out to the system by using their email and passwords.

3.1.1.2.1 Couriers

- Couriers can record the information of taking delivery from the customer into the system.
- They also should record the information of delivering the package to the destination customer into the system.
- When couriers can not find anyone at the destination address, the package is returned back to the branch.

3.1.1.2.2 Package Manager

- Package managers can create ship requests for the customer by taking necessary information from customers.

- Package managers can assign packages to vehicle drivers.
- Package managers can assign couriers to take packages from customer addresses.
- Package managers can assign couriers to deliver packages to customer addresses.
- Package managers can see, evaluate reports and give feedback to the customer about the state of the package.
- Package managers can refund according to the reports.
- In case of unsuccessful delivery of packages, package managers will record it as an undelivered package.

3.1.1.2.3 Shipper

- Shippers can record the information of taking delivery from the branch into the system.
- Shippers also should record the information of delivering the package to the destination branch into the system.

3.1.1.3 Administrator

- Administrators can log in and log out to the system by using their email and password.
- They are in control of employee profiles. They can create and remove employee profiles.
- They are able to update employee information.
- They can add or remove branches.

3.2 Non-Functional Requirements

3.2.1 Usability

3.2.1.1 User-Friendly Interface

- A help page will be provided for users to understand how to realize tasks.
- The elements of the webpage will be the optimum size to interact.
- The feedback system through the reports will be used to improve the experience of shipping.

3.2.1.2 Invalid Password or Email Address

- When a user enters the wrong password or email address, the system will not accept this information and show an error to the user.

3.2.2 Performance

- At least 1GB of data will be stored in the database.
- The delay for each operation will not exceed 2 seconds.

3.2.2 Security

- Time out can occur during the session of the user. In this case, to protect against certain types of computer attacks, the system will close the users' accounts.
- Employees and administrators must set a password with a minimum of 8 characters.

3.2.3 Reliability

- The system will be monitored by admins for error feedback.
- The system will be available for every moment (7/24).

3.3 Pseudo Requirements

- For the client-side of the application, Javascript will be used with React framework. For the server-side of the application, Node.js and Express.js will be used.
- For Structured Query Language (SQL), PostgreSQL will be used.

4. Limitations

4.1 User

- Emails of users will be unique.
- Every user must enter their name, surname, and email. However, phones will be optional.
- The phone numbers are restricted with 13 characters including country code.

4.1.1 Customer

- Every customer must enter their address including their city, street, apartment number, zip, state.
- For each customer card, the CVC code must be a minimum of 3 characters, and it is mandatory.

- For each customer card, the card number (c_id in E-R diagram) must be 16 characters and it is mandatory.
- For each order, customers have to enter the destination address.
- For each shipping request done by customers, at most 5 packages can be included.

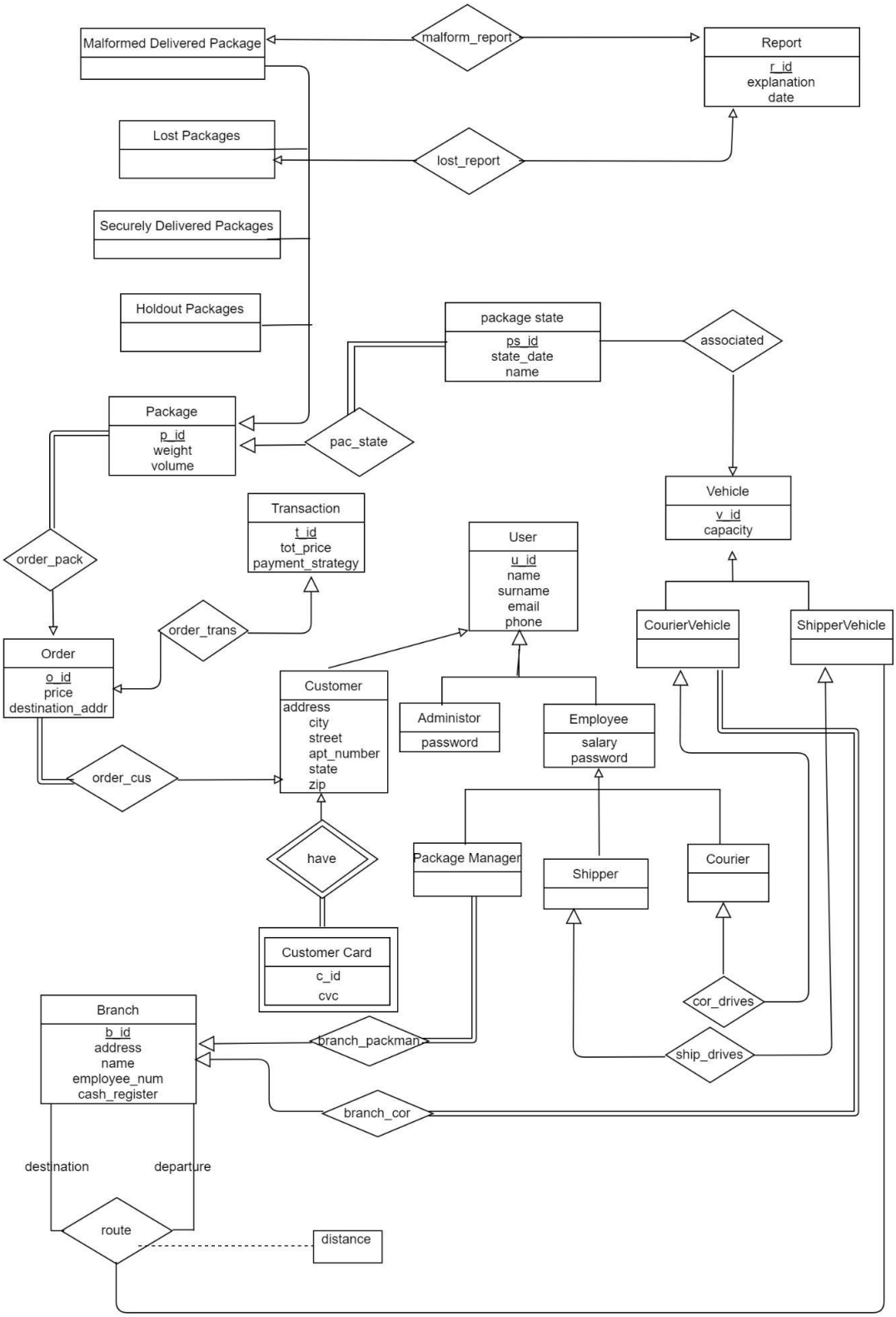
4.1.2 Employee

- For each administrator, it is essential to enter the address, the name of the new branch and the number of employees in that branch.
- For each package manager, it is essential to enter the weight and volume of the packages.

5. Conceptual Design (E/R Model)

Figure 1 demonstrates the Entity-Relationship Diagram of the project.

Figure 1 (ER Diagram)



6. Project Web Page

- https://ilke-kas.github.io/